

FVeinSyn: Synthetic Finger Vein Image Generator

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Abstract—A major challenge in finger vein recognition is the lack of large-scale public datasets. Existing datasets contain few identities and limited samples per finger, restricting the advancement of deep learning-based methods. To address this, we propose FVeinSyn, a large-scale controllable synthetic data generation framework for finger vein. It explicitly decouples synthesis of vascular topology and imaging appearance to mitigate the limitations caused by insufficient training samples, such as inadequate identity diversity and restricted realism. Specifically: first, a finger vein identity generator models vascular topology under physiological and geometric constraints using stochastic L-systems, producing anatomically valid and identity-distinctive vascular patterns. Then, a cascaded region-aware GAN renders the topological maps into realistic near-infrared images. Finally, an intra-class diversity generator introduces geometric and optical perturbations to simulate realistic intra-class variations. Using FVeinSyn, we generated 500,000 images (10,000 vein identities, 50 samples per identity) and conducted extensive evaluations. Results show that FVeinSyn holds significant advantages in realism, identity diversity, vascular pattern consistency, and intra-class diversity. Models trained with FVeinSyn outperform real-data-only baselines a cross eight public datasets, achieving an average accuracy improvement of 27.43%. The code is available at: <https://github.com/EvanWang98/Synthetic-Finger-Vein-Generator>.

Index Terms—Synthetic finger vein generation, biometrics, lindenmayer systems, adversarial generation, finger vein recognition

I. INTRODUCTION

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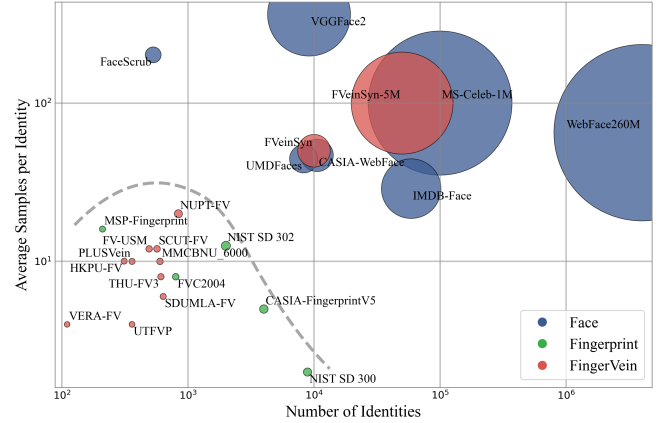


Fig. 1. Comparison of publicly available datasets representing face [5], [11]–[16], fingerprint [17]–[21], and finger vein [22]–[31] biometric modalities. FVeinSyn-5M (50,000 identities/5,000,000 samples) [32] and FVeinSyn (10,000 identities/500,000 samples) are synthetic finger vein datasets proposed in this work.

OVER the past two decades, biometric recognition technologies have achieved remarkable progress and have been widely deployed in scenarios such as security authentication, financial payment, and public safety [1]. As the first-generation modalities, face [2], [3] and fingerprint [4] recognition have benefited from large-scale datasets [5]–[7] and deep representation learning, reaching industrial-level maturity. However, they also expose significant security and privacy [8], [9] concerns: facial images can be easily captured and misused in public scenarios, while fingerprints are permanently exposed and thus vulnerable to duplication and spoofing. In contrast, finger vein recognition leverages subcutaneous vascular patterns, offering intrinsic resistance to forgery and stronger privacy protection [10]. Consequently, it has been recognized as a representative modality of the second-generation biometrics and has already seen industrial adoption (e.g., products from Hitachi, Fujitsu, and Tencent).

Despite its advantages, the development of finger vein recognition is still constrained by the severe lack of large-scale public datasets [33]. Unlike face and fingerprint images, finger vein acquisition requires active participation and a more complex capture process, making it difficult to collect large-scale samples from the Internet [5]. As a result, current finger vein datasets fall far behind face and fingerprint datasets in terms of identity scale, per-identity sample size, and intra-class diversity (see fig. 1). For instance, while face datasets now include over 4M unique identities and 260M images—enabling the success of deep representation learning—and fingerprint datasets reach nearly 10K identities, the largest available

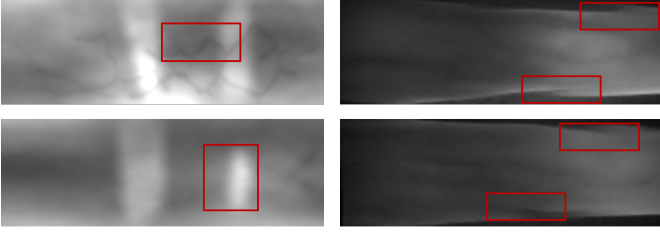


Fig. 2. Traditional modeling (left): vessels are unsmooth and rigid, with unrealistic infrared appearance. Learning-based (right): noticeable artifacts are observed.

finger vein dataset still contains fewer than 1K identities. This disparity directly hinders research on critical issues such as million to billion scale retrieval, cross-device generalization, and robustness evaluation.

Publicly available finger vein datasets include SDUMLA-FV [22], UTFVP [23], MMCBNU_6000 [24], FV-USM [25], HKPU-FV [26], PLUS-FV3 [27], SCUT-FV3 [28], THU-FV3 [29], VERA-FV [30], and NUTP-FV [31]. While they have provided valuable support for research, several limitations remain: (1) the number of unique identities is limited—fewer than 1K even in the largest dataset; (2) per-identity samples are scarce, typically ranging from 2 to 12 impressions per finger; (3) intra-class variations are minimal, as samples collected within the same session show almost no variation, with at most two sessions per dataset; and (4) auxiliary annotations such as joint cavity positions, ROI bounding box, vein pattern, and finger shape are unavailable, restricting the use of advanced deep learning techniques.

In various domains such as face and fingerprint recognition, synthetic datasets have been leveraged to improve model generalization under limited real data. Generative adversarial networks (GANs) [6] and diffusion models [34]–[36] have demonstrated strong image synthesis capabilities, effectively expanding training dataset. However, finger vein synthesis is fundamentally more challenging than generic image generation, because a high-quality synthesizer must simultaneously support large-scale identity creation, realistic intra-class variation simulation, and stable identity-label consistency. More importantly, unlike prior synthetic finger vein generation methods that either rely on explicit modeling of vascular geometry and optical priors or directly learn image-level mappings from limited real data, the proposed FVeinSyn adopts a structurally decoupled generation paradigm: vascular topology is generated explicitly in identity space, while near-infrared appearance is rendered separately in image space. This decoupling mechanism improves controllability and avoids forcing a single end-to-end generator to jointly learn complex vascular topology, finger geometry, and near-infrared imaging distributions from scarce real samples.

Traditional modeling of synthetic finger vein images primarily relies on explicit modeling methods based on geometric structures and optical priors [37]. Although these approaches can reproduce the overall vascular morphology, they often fail to generate smooth and physiologically coherent local details, resulting in vascular patterns that appear rigid or unnatural.

Moreover, empirical imaging models cannot fully capture the complex physical processes involved in near-infrared imaging, such as subsurface scattering and tissue absorption, leading to a gap in realism between synthetic and real images. Fig. 3 shows limitations in existing synthetic finger vein generation methods.

From a deep generative modeling perspective, data scarcity poses unique challenges to finger vein generation. The limited real samples cannot adequately cover the broad variability of vascular topologies, nor are they sufficient for a single end-to-end generator to simultaneously learn both the identity distribution (topology) and the appearance distribution (near-infrared imaging style) [38]. Moreover, generating sufficiently diverse vein patterns alone is already a challenging task; jointly modeling vein patterns, imaging characteristics, and finger geometry further increases the complexity. In particular, under limited data conditions, it remains extremely difficult to stably learn finger vein image samples with synthetically generated anatomically plausible finger geometries from scratch.

To address these challenges, we propose FVeinSyn, a large-scale and controllable synthetic finger vein image generation framework. FVeinSyn explicitly decouples identity generation (topology) and near-infrared appearance rendering, and further introduces controllable structural and degradation priors to model realistic complete finger vein images with rich inter-class and intra-class variations.

Specifically, we propose a vascular identity generator, which defines a stochastic vascular growth grammar constrained by anatomical and topological priors and implements it within a symbolic L-system framework. The generator models vascular formation as a recursive evolutionary process governed by probabilistic production rules, inspired by the biological mechanism of vascular bifurcation. By incorporating physiological constraints such as branching angles, diameter decay rates, and curvature smoothness during generation, the model produces vascular patterns that are both topologically continuous and physiologically plausible, overcoming the local rigidity and morphological discontinuity issues of traditional explicit modeling methods. Furthermore, it enables the synthesis of anatomically realistic binary vascular patterns in topological space with controllable and identity-distinctive priors.

We further propose a region-aware cascaded adversarial network as the synthetic image renderer. It takes binary vascular topology as input, allowing the generator to bypass the need to learn complex vascular topology distributions in image space and instead focus on modeling the near-infrared appearance style under the supervision of a limited number of real vein samples. Furthermore, we assign each vascular identity a unique set of anatomical structural priors—including finger shape, joint cavity regions, and regions of interest—to enhance identity distinctiveness. These structural priors can be obtained in a controllable and low-cost manner, making their integration both practical and scalable for large-scale synthesis. Guided by these priors, the renderer maps binary vein patterns to realistic near-infrared appearances. In addition, a cascaded regional supervision mechanism enforces multi-region consistency, thereby improving both the global realism and local fidelity of the synthesized images.

Finally, unlike existing synthetic finger vein generation methods that typically produce only localized vein-region images with subtle variations [37]–[41], FVeinSyn generates complete finger vein images and explicitly models intra-class diversity. To this end, we introduce an intra-class diversity generator, which incorporates geometric transformations (e.g., translation, rotation, and scaling), photometric changes (e.g., over/under exposure), and imaging degradations (e.g., motion blur, optical blur, and scattering blur) to more realistically simulate the complexity and diversity of real finger vein data.

The followings are the contributions of the paper.

- We propose FVeinSyn, the first large-scale, anatomically grounded, and controllable synthetic data generation framework for finger vein. It enables identity-consistent and realistic vein synthesis at scale without relying on large real samples.
- We propose a vascular identity generator that defines a novel stochastic vascular growth grammar constrained by finger physiological priors and implements it within a symbolic L-System formalism to generate anatomically plausible and identity-distinctive vascular topologies.
- We propose a region-aware cascaded renderer that maps binary vascular topologies into realistic near-infrared images through multi-region adversarial generation.
- We propose an intra-class diversity generator that simulates realistic intra-class variability through controllable geometric, photometric, and imaging transformations.
- Extensive experiments demonstrate that FVeinSyn-generated datasets improve downstream recognition performance, outperforming existing state-of-the-art finger vein generation methods in identity scale and diversity.

Our earlier conference work [32] released FingerVeinSyn-5M, the largest synthetic finger vein dataset to date, which was generated using FVeinSyn. That work primarily focused on the construction of the dataset and its preliminary evaluation. Compared with the conference version, this study provides a detailed description of the complete FVeinSyn synthetic finger vein image generation framework, elaborates on the design and implementation of its key components, including the Vascular Identity Generator, Region-Aware Cascaded Renderer, and Intra-Class Diversity Generator, and presents a significantly expanded evaluation of the generation framework.

II. RELATED WORK

A. Finger Vein Recognition

Finger vein recognition initially relied on handcrafted feature extraction methods, which use manually designed pattern descriptors and matching algorithms to classify vein identifies. Representative methods include repeated line tracking (RLT) [42], maximum curvature (MC) [43], Gabor filtering [27], isotropic undecimated wavelet transform (IUWT) [44], and wide line detector (WLD) [45]. Although these traditional methods can achieve basic recognition performance in controlled environments, their effectiveness is limited by insufficient feature representation and poor adaptability to environmental changes. They are sensitive to variations in

lighting, finger misalignment, and imaging noise [46]. The choice of matching strategy also affects performance.

With the development of deep learning, end-to-end training methods based on convolutional neural network have significantly improved recognition accuracy and system robustness, driving a paradigm shift in the field [47], [48]. Current deep learning approaches can be categorized into several aspects: network architecture design [49], [50], loss functions [51], prior knowledge [33], preprocessing [10], domain adaptation and robustness.

Firstly, for the architecture, researchers have gradually moved from shallow convolutional network to deeper and more expressive backbones, such as residual network [33], lightweight [52], and multi-scale branches [53], [54]. To address the inherently low contrast and fine pattern of finger veins, network designs often incorporate feature pyramids, full view [55] and attention mechanisms [56], [57] to enhance the response and localization of small vascular patterns. At the same time, the demand for lightweight and efficient inference has led to the development of compact network and pruning/quantization strategies that can be deployed on resource-limited devices [52].

Secondly, in terms of loss functions and training paradigms, metric learning approaches (contrastive loss, triplet loss, center loss) [58], [59] and angular/margin-based losses (ArcFace, CosFace) [51] have become mainstream methods for improving discriminative feature representation, significantly increasing inter-class separation and intra-class compactness. Recently, self-supervised [60] and contrastive learning [38], [61] methods have been applied to finger vein recognition to learn robust features. Meta learning [28] have also been explored for sample limited recognition.

Despite significant improvements on several benchmarks, existing deep learning methods still face performance limitations [62]. First, the scale and diversity of available data are insufficient; the number of identities, the number of samples per identity, and the number of acquisition sessions in public datasets are far from meeting the requirements for learning broad distributions in deep models. Second, intra-class variations are complex, with posture, light scattering, partial occlusion, and differences in imaging devices all contributing to high-dimensional variations that simple augmentation and end-to-end training cannot fully capture [63]. These challenges directly limit the model's generalization ability in cross-domain, few-shot, and large-scale retrieval scenarios [64].

B. Synthetic Finger Vein Generation

In recent years, the generation of synthetic biometric data has become an important direction for enhancing the generalization and scalability of recognition models. In the field of finger vein recognition, existing studies can be broadly divided into two categories: intra-class generation [33], [64] and inter-class generation [32], [37], [38]. The former aims to produce diverse supplementary samples for known identities to improve intra-class robustness, while the latter focuses on constructing new identities along with their corresponding multi-modal samples to expand the identity space and support

model pretraining. Currently, most research still concentrates on intra-class generation, with only a few studies attempting to address the inter-class generation problem.

Hillerström *et al.* [37] first explored an analytical modeling-based approach for synthesizing finger vein images and released a synthetic dataset containing about 5,000 identities, each with 10 samples. This pioneering study laid the foundation for subsequent research, but it still showed clear limitations in visual realism, geometric continuity, and intra-class diversity, as its variations were mainly introduced through limited parameter perturbations. Later, Ou *et al.* [38] proposed an image generation framework based on ROI block shuffling and generative adversarial network, which achieved more natural pattern appearance through region remapping and adversarial rendering. However, due to insufficient training samples, most of the generated sample contain artifacts.

In contrast, related biometric modalities such as palm vein and palmprint synthesis have achieved more systematic progress, with approaches based on morphological modeling, cross-modal style transfer, and conditional generation [35], [39]–[41], [65]. Nevertheless, the synthesis of finger vein images remains more challenging. First, it requires accurate reconstruction of fine anatomical structures of real fingers (such as joint cavities and finger shape). Second, the imaging process is affected by complex optical effects, including near-infrared light transmission, subsurface scattering, and local occlusion—factors that make generation far more difficult than for palm or dorsal hand regions. Existing research mostly focuses on local ROI-level synthesis, whereas high-fidelity, full-finger generation has greater potential for identity modeling, structural prior learning, and multi-task recognition [37], [38].

C. Biometrics with Synthetic Dataset

Compared with real biometric data collected from individuals, synthetic data offers significant advantages in terms of ethical compliance and distributional balance [66]. In recent years, large-scale real datasets have raised growing concerns regarding privacy protection, lack of informed consent, and racial bias [34]. In contrast, synthetic data can be expanded at scale without violating personal privacy, effectively mitigating such risks. Moreover, the generation process of synthetic samples can be strictly controlled, allowing researchers to precisely balance factors such as identity distribution, gender ratio, and imaging conditions, thereby reducing the impact of class imbalance on model learning [67].

Nevertheless, recognition systems trained solely on synthetic data still face performance limitations. The main reason lies in the texture fidelity and domain distribution gap of generated samples, which restrict their generalization capability to real-world test sets. Therefore, in most biometric recognition tasks, synthetic data is typically used as a supplement to real data. In face recognition [34], [68], [69], fingerprint recognition [4], [6], [36], [70], palmprint recognition [35], [40], [41], [65], and anti-spoofing tasks [71]–[73], numerous studies have demonstrated that joint training with real and synthetic samples can effectively improve both recognition accuracy and robustness.

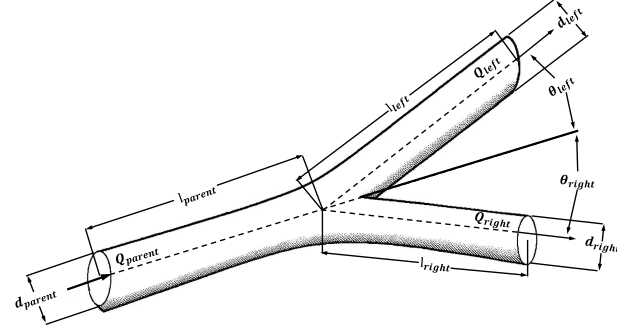


Fig. 3. **Bifurcation parameters:** diameters (d_{parent} , d_{left} , d_{right}), lengths (l_{parent} , l_{left} , l_{right}), volumetric flows (Q_{parent} , Q_{left} , Q_{right}) and bifurcation angles (θ_{left} , θ_{right}).

This trend indicates that the value of synthetic data is shifting from simple data augmentation toward task-consistent enhancement—that is, improving the generalization boundary of feature representation through complementary distributions with real samples. Consequently, high-fidelity, controllable, and task-aligned synthetic data is expected to become a key foundation for advancing the next generation of biometric recognition systems.

III. PROPOSED APPROACH

Our proposed framework synthesizes realistic finger vein images through a three stage process. Fig. 4 illustrates the overview of the FVeinSyn framework. First, a binary vascular topology map I_P is generated, which recursively simulates the branching of finger veins under physiological and geometric constraints. Next, the generated topology map I_P is fed into a cascaded region-aware generative adversarial network (CascadedRA-GAN) along with a set of structural priors $\mathcal{S}$, including finger shape masks and joint-cavity annotations. This module renders I_P into a realistic near-infrared image I_{syn} . Finally, to further enhance intra-class diversity, the intra-class diversity generator (ICDG) applies a sequence of geometric and optical perturbations in a probabilistic manner, simulating pose variations, skin scattering, exposure changes, and imaging blur. Thus, by FVeinSyn framework can synthesize a wide range of realistic finger vein images for diverse identities and acquisition conditions. Each of these components is described in detail in the subsections below.

A. Preliminary

Vascular Network. The topology and geometry of vascular network can be characterized by a set of fundamental physical-geometric constraints, primarily involving flow conservation and a power-law relationship of branch radii [74], [75]. At a bifurcation, let the volumetric flows of the parent vessel and the left and right child vessels be Q_{parent} , Q_{left} , Q_{right} , respectively, and their radii be d_{parent} , d_{left} , d_{right} . The branching asymmetry ratio is defined as $\alpha = d_{\text{right}}/d_{\text{left}}$, and the left and right child branches form inclination angles θ_{left} , θ_{right} relative to the parent segment, as illustrated in Fig. 3.

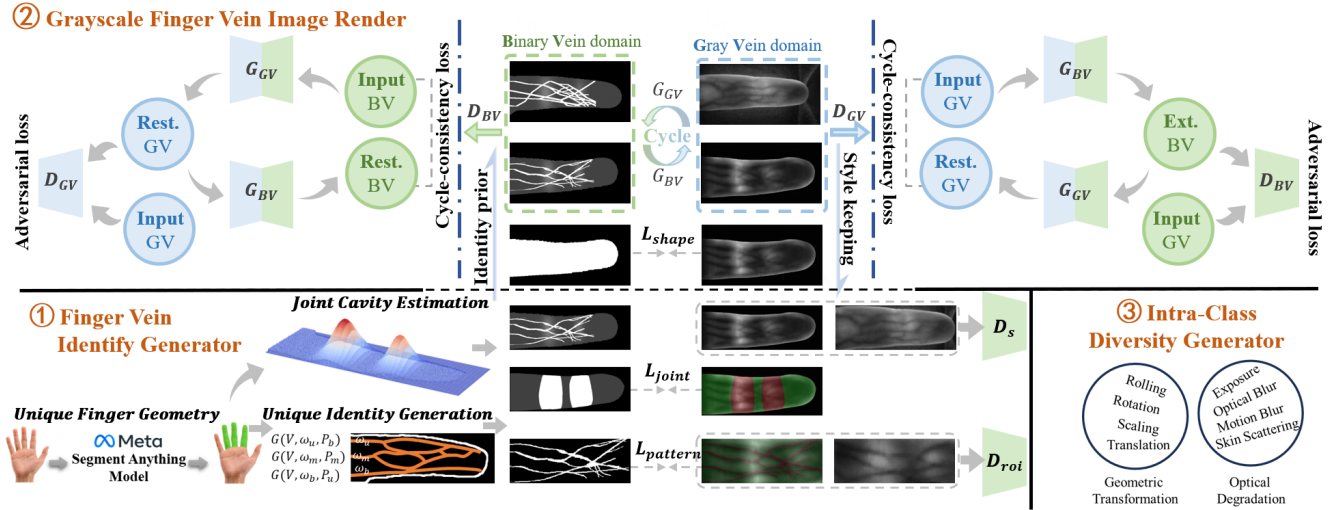


Fig. 4. Overview of the FVeinSyn framework for large-scale synthetic finger vein image generation. 1) Finger Vein Identity Generator, where unique vascular topologies are created. 2) Grayscale Finger Vein Image Render, performed by a Cascaded Region-Aware GAN (CascadedRA-GAN) to produce realistic images. 3) Intra-Class Diversity Generation, which applies geometric and optical perturbations to enhance variability within each identity.

Flow Conservation requires that the incoming flow equals the total outgoing flow:

$$Q_{\text{parent}} = Q_{\text{left}} + Q_{\text{right}}, \quad (1)$$

meanwhile, vessel radii satisfy a *Power-Law Constraint*, commonly referred to as the Murray-type law:

$$d_{\text{parent}}^\gamma = d_{\text{left}}^\gamma + d_{\text{right}}^\gamma \quad (2)$$

where the branching exponent γ is typically in the range 2.55–3.0, with the classical case $\gamma \approx 3$ corresponding to Murray's law [76]. The asymmetry ratio α quantifies the imbalance between the two child branches.

During vascular tree generation or modeling, each vessel segment grows until it reaches a threshold length l , at which point it bifurcates into two child segments oriented by angles θ_{left} and θ_{right} . These constraints jointly ensure the consistency of blood flow and geometric scaling in vascular network. **Lindenmayer Systems** (L-systems) are formal grammar systems that provide a mathematical framework for modeling the growth processes of organisms, particularly plants [77]. A context-free L-System (OL-System) applies productions without considering the context of symbols within a string. Formally, it is defined as a triple $L = (V, P, \omega)$, where V is a finite set of symbols (alphabet) that may include terminal and non-terminal symbols; P is a finite set of production rules, each of the form $(A, \alpha) \in P$, denoted as $A \rightarrow \alpha$, where A is the predecessor and α is the successor string; and $\omega \in V^*$ is the axiom, serving as the initial string for the derivation process. Geometrically, OL-Systems are typically interpreted by mapping symbols to drawing instructions, thereby generating complex structures such as fractals or branching patterns. For instance, a symbol may represent moving forward in a given direction or rotating by a specified angle in the drawing plane.

When each symbol A has exactly one associated production rule, the OL-System is referred to as deterministic (DOL-System). Conversely, if a function assigns a probability to each

production, the resulting system is a context-free stochastic L-System (SOL-System), which is the focus of this work [78]. In SOL-Systems, probabilities are introduced to production rules, rendering the derivation process non-deterministic. Formally, the system is represented as a quadruple $L = (V, P, p, \omega)$. For any predecessor symbol A , the sum of probabilities of all corresponding productions equals one,

$$\sum_{A \rightarrow \alpha \in P} p(A \rightarrow \alpha) = 1. \quad (3)$$

The probabilistic mechanism of SOL-Systems introduces randomness into the derivation process. In addition, production rules can be parameterized to simulate natural variations in growth patterns, such as adjusting branching angles and segment lengths, enabling the generation of realistic synthetic vascular network.

Adversarial Image-to-Image Style Transfer. Let X and Y denote two image domains with real data distributions p_X and p_Y , respectively. Adversarial image-to-image style transfer aims to learn a pair of mappings $G : X \rightarrow Y$, $F : Y \rightarrow X$, along with discriminators D_Y and D_X that distinguish whether samples belong to the target domain. The main goal of this class of methods is to perform cross-domain image translation while preserving the semantic and geometric content of the input images, often without requiring paired training samples [79].

To ensure that the generated images $G(x)$ follow the target distribution p_Y , an adversarial loss is defined as

$$\mathcal{L}_{\text{GAN}}^{X \rightarrow Y} = \mathbb{E}_y[(D_Y(y) - 1)^2] + \mathbb{E}_x[D_Y(G(x))^2]. \quad (4)$$

Since relying solely on adversarial loss may result in degenerate mappings or content loss, a cycle-consistency constraint is introduced:

$$\mathcal{L}_{\text{cyc}} = \mathbb{E}_{x \sim p_X}[\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_Y}[\|G(F(y)) - y\|_1]. \quad (5)$$

This encourages $F \circ G \approx \text{id}_X$ and $G \circ F \approx \text{id}_Y$, thereby preserving input content and enabling invertible mappings.

B. Finger Vein Identity Generator

Finger vein patterns exhibit inherently individual-specific vascular structures, whose branching topology and spatial distribution remain highly stable across multiple acquisitions, yet differ significantly between individuals. Features such as bifurcation geometry, vessel diameter ratios, and directional organization form the structural foundation of biometric recognition. Therefore, the synthesis of vein identities must balance anatomical plausibility and topological diversity, ensuring that each identity corresponds to a physiologically valid yet unique vascular pattern.

To this end, we model vascular formation as a novel probabilistic growth grammar whose production rules correspond to biologically meaningful branching events, where rule probabilities are parameterized by anatomical and geometric priors of the finger, implementing it within a symbolic L-System framework to generate anatomically plausible and identity-distinctive vascular topologies. Specifically, the generation starts from an initial trunk at the finger base, and child branches are recursively created according to stochastic production rules. The grammatical derivation produces a symbolic string, in which each symbol is mapped to a local drawing primitive—such as “move forward by length l ,” “rotate by angle θ ,” or “generate branch.” Sequential execution of these primitives yields a set of 2D line-segment coordinates, known as *turtle trajectories*. Bézier interpolation is then applied to these trajectories to form smooth and continuous vascular curves, which are subsequently rasterized into a binary image I_P , where the vessel thickness is determined by the average diameter of each segment.

We describe the generation process from the perspective of a single vascular trunk and its recursive bifurcations. Beginning with an initial segment of length l_0 and diameter d_0 , the model grows recursively following parameterized stochastic production rules. For a parent vessel $F(l)$ of length l . At each recursion step, a branching production rule is applied with probability $p \in (0, 1)$, where p is a fixed model parameter controlling local branching frequency. A typical production rule can be expressed as:

$$F(l) \rightarrow p[+\theta F(l \cdot \eta_l)][-\theta F(l \cdot \eta_l)], \quad (6)$$

where $\eta_l \in (0, 1)$ denotes the inter-level length attenuation factor and θ represents the branching angle. The bracket operators $[\cdot]$ indicate push and pop operations on the growth state stack, which stores the current position and orientation. With probability $1 - p$, the vessel continues elongation without branching. Here, $\eta_l \in (0, 1)$ denotes the inter-level length attenuation factor. The vessel diameter is attenuated by a scaling factor η_d at each growth step, which is chosen to approximately satisfy the Murray-type law [76].

Through this process, the generator produces vascular network with clear hierarchical organization and diverse topological structures. Each synthetic identity typically contains three to four primary trunks (distributed vertically along the finger axis) and their secondary branches, exhibiting physiologically consistent asymmetry and density distributions. Finally, the generated vascular structure is rasterized into a binary *topology*

map I_P , serving as the foundational representation for finger vein identity generation and providing anatomically consistent, controllable priors for the subsequent image rendering stage.

C. Finger Vein Image Renderer: CascadedRA-GAN

The synthesis of realistic finger vein images can be formally described as a conditional generative modeling problem, where the objective is to map a binary vein identity pattern I_p into a realistic near-infrared image while preserving both anatomical plausibility and inter-subject variability. To this end, we propose a cascaded region-aware generative adversarial network (CascadedRA-GAN), whose central idea is to decompose the synthesis process into two coupled sub-tasks: (i) mapping binary vein patterns to realistic appearance under structural priors; and (ii) enforcing multi-region consistency through cascaded regional supervision, thereby enhancing global realism and local fidelity simultaneously.

Specifically, given a binary vein identity pattern I_p , the generator further takes as input a set of structural constraints:

$$\mathcal{S} = \{I_s, D_j, A\}, \quad (7)$$

where I_s denotes the finger shape mask, D_j represents the brightness distribution of joint cavities, and A comprises auxiliary annotations including joint cavity coordinates, region of interest, and local deformation descriptors. The rendering task can thus be formulated as learning the mapping:

$$G : (I_p, \mathcal{S}) \mapsto I_{\text{syn}}, \quad (8)$$

where I_{syn} denotes the synthesized finger vein image.

In contrast to approaches relying on approximate or hand-crafted priors, our design explicitly incorporates real anatomical structures as generative constraints. Concretely, we leverage a large-scale gesture database [80] with the *Segment Anything Model* [81] to achieve precise segmentation of finger regions, thereby producing explicit masks for ROI, joint cavities, and overall finger. This ensures that each synthesized vein identity is embedded within a unique and anatomically consistent finger structure, effectively preventing mode collapse and enhancing both inter-subject diversity and realism.

To balance global consistency with local structural fidelity, we design a cascaded region-aware loss for the proposed CascadedRA-GAN. The overall objective is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{CycleGAN}} + \mathcal{L}_{\text{Region-Aware}}, \quad (9)$$

where $\mathcal{L}_{\text{CycleGAN}}$ includes standard identity, cycle-consistency, and adversarial losses to ensure distributional alignment between synthesized and real samples at global scale.

The region-aware loss $\mathcal{L}_{\text{Region-Aware}}$ is defined as

$$\mathcal{L}_{RA} = \mathcal{L}_{\text{cycle}}^s + \mathcal{L}_{\text{cycle}}^{\text{roi}} + \mathcal{L}_{\text{adv}}^s + \mathcal{L}_{\text{adv}}^{\text{roi}} + \lambda_1 \mathcal{L}_s + \lambda_2 \mathcal{L}_j + \lambda_3 \mathcal{L}_p, \quad (10)$$

where $\mathcal{L}_{\text{cycle}}^s$ and $\mathcal{L}_{\text{cycle}}^{\text{roi}}$ denote cycle-consistency constraints imposed on the finger-shape region and the ROI, respectively, while $\mathcal{L}_{\text{adv}}^s$ and $\mathcal{L}_{\text{adv}}^{\text{roi}}$ represent the corresponding regional adversarial losses. These local discriminators operate independently of the global discriminator, thereby capturing fine-grained style variations at multiple scales.

TABLE I
PARAMETERIZATION FOR ICDG OPERATORS. $\mathcal{U}(a, b)$ DENOTES THE UNIFORM DISTRIBUTION OVER THE INTERVAL $[a, b]$. W, H ARE WIDTH AND HEIGHT, τ IS THE TRANSLATION FACTOR.

Operator	Params	Range / distribution	Default	Notes
Translation	t_x	$U(-\tau_w W, \tau_w W)$	$\tau_w = 0.05$	horizontal displacement vertical displacement
	t_y	$U(-\tau_h H, \tau_h H)$	$\tau_h = 0.05$	
Rotation	ϕ	$U(-\phi_{\max}, \phi_{\max})$	$\phi_{\max} = 15^\circ$	rotation angle
Rolling	ψ	$U(-\psi_{\max}, \psi_{\max})$	$\psi_{\max} = 10^\circ$	rolling angle
Scaling	s	$U(1 \pm s_{\max})$	$s_m = 0.1$	scale factor
Exposure	v_0	$U(-v_{\max}, v_{\max})$	$v_{\max} = 2$	exposure intensity
Skin scatter	β	$U(\beta_{\min}, \beta_{\max})$	$\beta_{\min} = 0.05, \beta_{\max} = 0.10$	scattering coefficient decay rate
	α	$U(\alpha_{\min}, \alpha_{\max})$	$\alpha_{\min} = 0.02, \alpha_{\max} = 0.06$	
Motion blur	L	$Choice\{L_i\}$	$\{L_i\} = \{1, 2, \dots, 15\}$	kernel length blur direction
	θ	$U(0, 2\pi)$	$\theta_{\max} = 2\pi$	
Optical blur	K	$Choice\{K_i\}$	$K_i = \{3, 5, 7, 9, 11, 15\}$	kernel size standard deviation
	σ	$U(0, \sigma_{\max})$	$\sigma_{\max} = 10$	
Trigger ($\mathcal{G} / \mathcal{O}$)	ω_k	weights ω_k	$\omega_{\text{trans}} = 0.8, \omega_{\text{rot}} = 0.8, \omega_{\text{roll}} = 0.5, \omega_{\text{scale}} = 0.5,$ $\omega_{\text{expo}} = 0.3, \omega_{\text{scatter}} = 0.8, \omega_{\text{motion}} = 0.5, \omega_{\text{opt}} = 0.5$	trigger probability

In addition, the shape-consistency loss $\mathcal{L}_s$ is based on the binary finger-shape mask I_s , where Hausdorff distance (Haber loss) is employed to constrain the contour alignment. The joint-cavity loss $\mathcal{L}_j$ and the vein-pattern loss $\mathcal{L}_p$ are inspired by physiological priors of near-infrared imaging. Specifically, joint-cavity regions generally exhibit higher average brightness, whereas vein regions appear darker. To model this explicitly, we introduce a contrastive loss:

$$\mathcal{L}_{j/p} = \frac{1}{N} \sum_{i=1}^N \max(0, \text{Margin} - (\mu_A(x_i) - \mu_B(x_i))), \quad (11)$$

where x_i denotes the i -th generated image, and $\mu_A(\cdot) / \mu_B(\cdot)$ indicate average pixel intensity within regions A and B , respectively. In $\mathcal{L}_j$, region A denotes the finger joint cavity region obtained from D_j , while region B denotes the finger mask region obtained from I_s . In $\mathcal{L}_p$, region B corresponds to the vein pattern region obtained from I_p , and region A represents the non-vein region, as illustrated in Fig. 4. For joint cavities and vein regions, we set $\text{Margin}_j = \text{Margin}_p = 0.2$, thereby enforcing explicit discriminability in brightness distributions across tissue types.

This multi-level loss design ensures that the generator not only learns global distributional consistency but also captures fine-grained structural, brightness, and pattern constraints, thereby enhancing both realism and discriminability in cross-identity synthesis.

D. Intra-Class Diversity Generator

To enhance the intra-class diversity of finger vein image samples, we design a two-stage intra-class diversity generator (ICDG). Given an identity topology map I_p and a structural prior $\mathcal{S}$, ICDG first applies geometric perturbations in the binary topology domain to simulate variations in pose and viewpoint. Subsequently, optical degradations are introduced in the grayscale rendering domain to mimic realistic near-infrared acquisition conditions, such as exposure, skin scat-

tering, and imaging blur. Table I summarizes the parameter settings for each ICDG operator.

We define a composite transformation associated with a transformation family P , which can be either the geometric family $\mathcal{G}$ or the optical family $\mathcal{O}$. The composite mapping is constructed as an ordered function composition of a subset of elementary transformations and is defined as

$$T_P(\cdot; \theta_P) = T_{k_n}(\cdot; \theta_{k_n}) \circ T_{k_{n-1}}(\cdot; \theta_{k_{n-1}}) \circ \dots \circ T_{k_1}(\cdot; \theta_{k_1}), \quad k_i \in P_{\text{act}}. \quad (12)$$

where $T_k(\cdot; \theta_k)$ denotes the k -th elementary transformation parameterized by θ_k . $P_{\text{act}} = \{k_1, \dots, k_n\} \subseteq P$ represents the activated subset of transformations selected according to the triggering strategy, and $\theta_P = \{\theta_{k_i}\}_{i=1}^n$ denotes the corresponding set of transformation parameters. So, the ICDG pipeline can be expressed as

$$\tilde{I}_p = T_{\mathcal{G}}(I_p; \theta_{\mathcal{G}}), \quad I_{\text{syn}} = G(\tilde{I}_p, \mathcal{S}), \quad \hat{I} = T_{\mathcal{O}}(I_{\text{syn}}; \theta_{\mathcal{O}}), \quad (13)$$

where $\mathcal{G} = \{\text{translation}, \text{rotation}, \text{rollnig}, \text{scaling}\}$ and $\mathcal{O} = \{\text{exposure}, \text{skin scatter}, \text{motion blur}, \text{optical blur}\}$. Based on the above formulation, we further define the parameterized sub-transformations $T_k(\cdot; \theta_k)$ for geometric perturbations and optical degradations as follows.

1) Geometric Perturbations (Binary/Topology Stage):

Translation: $T_{\text{trans}}(I; (t_x, t_y))$ will translate the finger within the plane, $t_x \sim U(-\tau_w W, \tau_w W)$, $t_y \sim U(-\tau_h H, \tau_h H)$.

Rotation: $T_{\text{rot}}(I; \phi)$ rotates the finger around its center within the plane, the angle sampling set to $\phi \sim U(-\phi_{\max}, \phi_{\max})$.

Rolling: $T_{\text{roll}}(I; \psi)$ simulates rolling effect of the finger by rotating around its longitudinal axis. Based on two assumptions: the cross-section of the finger is approximately elliptical, and the imaged veins are close to the finger's surfac, we employ an elliptical normalization scheme. Let (u, v) denote the normalized elliptical coordinates:

$$u = \frac{x - x_c}{a}, \quad v = \frac{y - y_c}{b}, \quad (14)$$

where x_c and y_c are finger center, a and b are the semi-major and semi-minor axes of the finger region. The rolling transform applies an angular offset

$$\Delta\theta(u, v) = \psi \cdot \sqrt{u^2 + v^2}, \quad (15)$$

where $\psi \sim U(-\psi_{\max}, \psi_{\max})$ controls the maximum rolling degree. The transformed coordinates are mapped as

$$I'(x', y') = R_{\Delta\theta(u, v)}(x, y), \quad (16)$$

with $R_{\Delta\theta}$ denoting a local rotation operator centered at (x_c, y_c) .

Scaling: $T_{\text{scale}}(I; s)$ uniformly scales fingers by a scaling factor s .

2) **Optical Degradation (Grayscale/Render Stage):**

Over/Under Exposure: $T_{\text{expo}}(I; (c_x, c_y), v_0)$ introduces local exposure through a centrally symmetric, exponentially decaying exposure field. (c_x, c_y) is a random center sampled within the finger region and $d(x, y) = \|(x, y) - (c_x, c_y)\|_2$. Denote $d_{\max} = \max_{(x, y) \in \text{ROI}} d(x, y)$. v_0 denotes the maximum exposure exponent. The exposure exponent field is

$$e(x, y) = v_0 - \delta d(x, y), \quad \delta = \frac{v_0}{d_{\max}}, \quad (17)$$

and the multiplicative gain is $f(x, y) = 2^{e(x, y)}$. The operator applies

$$I'(x, y) = \text{clip}(f(x, y) I(x, y), 0, 255). \quad (18)$$

Skin Scattering: $T_{\text{scatter}}(I; \alpha, \beta, A_{bm})$ employs an atmospheric scattering model to simulate skin subsurface scattering and haze effects in the near-infrared spectrum.

The spatially varying transmittance is defined as

$$t(x, y) = \exp(-\beta d_s(x, y)), \quad (19)$$

and the scattered image is computed by

$$I'(x, y) = I(x, y) t(x, y) + A_{bm}(1 - t(x, y)), \quad (20)$$

where $d_s(x, y)$ is a spatial distance proxy (e.g., $d_s(x, y) = \max(0, -\alpha \cdot \text{dist}(x, y) + C)$), α is a spatial decay factor, β is the scattering coefficient, and A_{bm} denotes the ambient light [82].

Motion Blur: $T_{\text{motion}}(I; L, \theta)$ simulates relative motion by convolving with a direction-dependent linear kernel $K_{L, \theta}$. Motion blur is simulated by convolution with a normalized linear kernel $K_{L, \theta}$:

$$I'(x, y) = (I(x, y) * K_{L, \theta})(x, y), \quad (21)$$

where $K_{L, \theta}$ denotes a discrete line segment kernel of length L along direction θ , normalized.

Optical Blur: $T_{\text{opt}}(I; \sigma)$ approximates the defocus or residual blur in an imaging system using a Gaussian point spread function (PSF):

$$I'(x, y) = (I * G_{\sigma})(x, y), \quad G_{\sigma}(r) = \mathcal{N}(0, \sigma^2), \quad (22)$$

3) **Triggering Strategy:** The ICDG adopts a two-stage probabilistic operator triggering strategy. Specifically, for each sample, the number of activated operators is first drawn, and then a subset of operators is selected from the operator family according to normalized weights without replacement.

Formally, the activation tendency of each operator $k \in \mathcal{P}$ is encoded by a non-negative base weight $\omega_k > 0$, which is normalized into a probability $p_k = \frac{\omega_k}{\sum_{j \in \mathcal{P}} \omega_j}$. The number of active operators is modeled as a Poisson random variable $K \sim \text{Poisson}(\lambda)$, truncated $K \leq K_{\max}$. This yields a long-tailed distribution, where most samples undergo only a few perturbations, while a small number of samples are subject to stronger distortions. This better reflects the intra-class variability observed in real finger vein imagery.

Finally, K operators are sampled without replacement from $\mathcal{P}$ according to $\{p_k\}$, forming the activated set $\mathcal{P}_{\text{act}}$.

IV. EXPERIMENT RESULTS

In this section, we conduct a comprehensive evaluation of the proposed FVeinSyn framework. We first assess the realism and quality distribution of generated images. Subsequently, quantitative recognition experiments are conducted to compare the performance of real and synthetic data under various scenarios, including open-set recognition, cross-domain recognition, and sample/identity-limited recognition. Finally, we analyze the synthetic dataset on identity uniqueness, vascular patterns consistency, and intra-class diversity.

A. Experimental Setup

Datasets: We utilized eight publicly available datasets: SDUMLA-FV [22], UTFVP [23], MMCBNU_6000 [24], PLUS-FV3 [25], FV-USM [26], HKPU-FV [27], SCUT-FV3 [28], and THU-FV3 [29] totaling 3938 identities and 35588 finger vein samples. During finger vein sample generation, since not all datasets contained complete finger regions, only HKPU-FV, FV-USM, and PLUS-FV3 were used for region supervision and shape supervision.

Generator Model Training Configuration: We generated 10,000 identities, each with 50 samples. The synthetic vein images had dimensions of 300×600 , while the finger-shaped vein images and finger vein ROI images were sized at 100×300 . We employed the Adam optimizer ($\beta_1 = 0.5$, $\beta_2 = 0.999$) for model training, with an initial learning rate of $2e-4$ that decayed after 50 epochs. The generator model was implemented using the PyTorch framework and trained on 4 NVIDIA RTX 4090 GPUs with a batch size of 24.

Recognition Model Training Configuration: The recognition model utilized a ResNet101 backbone [83] integrated with ArcFace [84] (scale factor $s = 64$, margin $M = 0.5$). The recognition model is firstly pretrained on synthesized data for 20 epochs and then finetuned on real datasets for 50 epochs. Finger vein ROI images sized at 100×300 , with a learning rate of 0.1 for training and $1e-2$ for fine-tuning, respectively. The model was implemented in PyTorch and trained on 4 NVIDIA RTX 4090 GPUs with a batch size of 256.

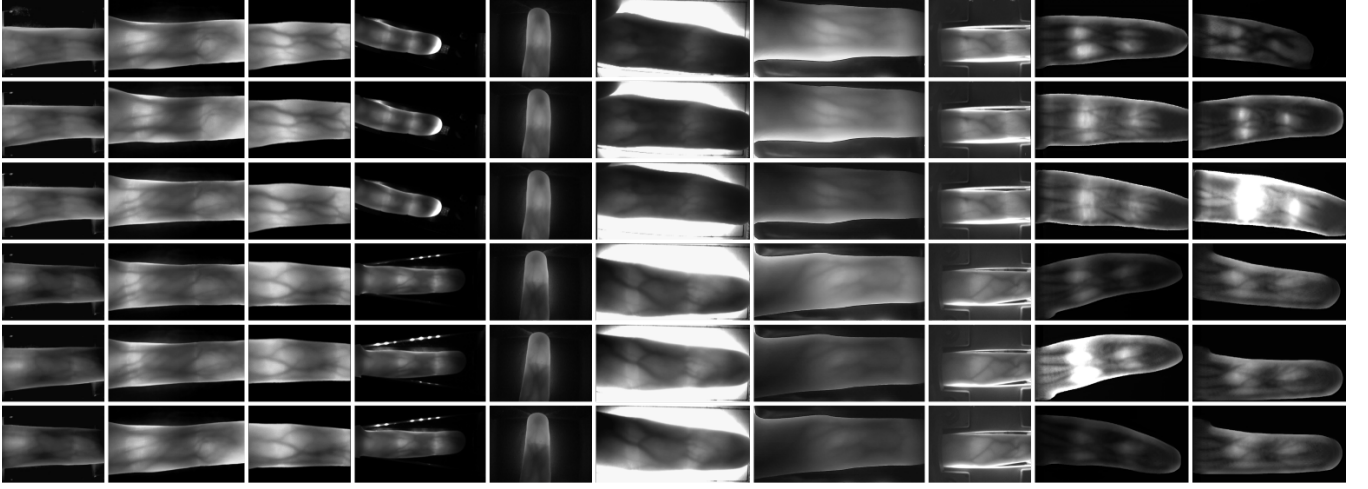


Fig. 5. Comparison between real and synthetic finger vein images generated by FVeinSyn. From left to right: SDUMLA-FV, UTFVP, MMCBNU_6000, PLUS-FV3, FV-USM, HKPU-FV, SCUT-FV3, THU-FV3, and FVeinSyn generate synthetic finger vein images (two columns).

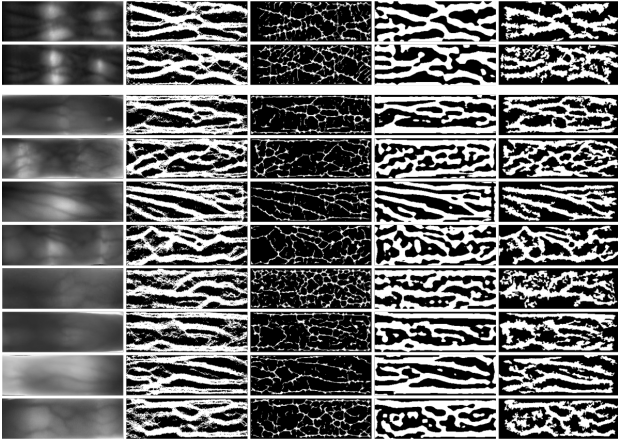


Fig. 6. Vein pattern extraction results on real and synthetic finger vein images using classical methods, including RLT, MC, Gabor filter, and IUWT. From top to bottom: FVeinSyn-generated synthetic finger vein images (two rows), SDUMLA-FV, UTFVP, MMCBNU_6000, PLUS-FV3, FV-USM, HKPU-FV, SCUT-FV3, and THU-FV3.

B. Visualization Analysis

To visually demonstrate the effectiveness of the FVeinSyn framework, we first present the generated images. Fig. 5 shows a comparison between real samples (the first eight columns) and FVeinSyn-generated synthetic samples (the last two columns), with each column containing three rows corresponding to different samples of the same subject. As shown, the synthetic samples exhibit a high degree of anatomical consistency: the main veins are clearly delineated, branching structures are coherent, and the overall topology is anatomically plausible. In terms of appearance, the synthetic images closely resemble real near-infrared acquisitions, exhibiting similar local illumination, scattering, and pattern details.

To further validate the similarity of the synthetic images at the pattern level, several classical vein pattern feature extraction methods were applied to both real and synthetic samples, including repeated line tracking [42], maximum curvature [43], Gabor filter [27], and isotropic undecimated

wavelet transform [44]. Fig. 6 presents the extraction results of these methods on both real and synthetic images. It can be observed that, in terms of pattern locations, line connectivity, and local response intensity, the pattern feature distributions of the synthetic images closely match those of the real images, indicating that FVeinSyn effectively preserves vein pattern structures and local details.

C. Image Quality Assessment

We further conducted a quantitative assessment of image quality across eight publicly available real finger vein datasets and three synthetic datasets. The experiments employed five reference-free image quality metrics: global contrast (GCF) [85], Entropy [86], Tenengrad, natural image quality evaluator (NIQE) [87], and blind image space quality evaluator (BRISQUE) [88]. Among these, GCF reflects the separability between the vein pattern region and surrounding tissue, entropy characterises the information richness of the vein pattern, and the Tenengrad gradient is used to assess the sharpness of the generated image. NIQE and BRISQUE evaluate the overall visual quality of the image from the perspectives of naturalness and perceptual distortion, respectively.

Table II lists the average scores across the five metrics for each dataset. Fig. 7 further illustrates the score distributions for real and synthetic datasets across the three key metrics: GCF, entropy, and Tenengrad. The interpretation of these metrics is closely related to their underlying physical meanings and the characteristics of real datasets. Specifically, GCF measures global image contrast and reflects the discriminability between vein patterns and surrounding tissues. Higher GCF values indicate clearer and more distinguishable vein structures.

As shown in Fig. 6, the UTFVP and MMCBNU_6000 datasets exhibit relatively clear vein structural characteristics, and accordingly present higher GCF values. In comparison, the GCF distribution of images generated by FVeinSyn is closer to that of these high-quality real datasets. Similarly, entropy reflects the information representation characteristics and statistical complexity of vein patterns within the region of

TABLE II
COMPARISON OF IMAGE QUALITY METRICS ACROSS DIFFERENT FINGER VEIN DATASETS.

Metrics	Real Datasets									Synthetic Datasets		
	SDUMLA-FV	UTFVP	MMCBNU_6000	PLUS-FV3	FV-USM	HKPU-FV	SCUT-FV	THU-FV3	Avg.	Ou <i>et al.</i> [38]	BTUSyn [37]	FVeinSyn(Our)
GCF	2.02	2.50	2.43	2.26	1.35	2.73	1.99	2.33	2.16	1.64	1.95	2.40
Entropy	6.62	7.16	6.84	6.33	6.08	6.88	6.44	6.58	6.64	6.45	6.67	7.06
Tenengrad	18.96	26.31	11.79	26.92	7.96	19.83	11.87	25.26	18.73	15.61	13.09	24.12
NIQE	30.28	31.18	32.74	24.84	31.86	31.53	31.53	33.27	31.16	27.02	29.90	28.59
BRISQUE	53.11	46.09	67.52	57.56	61.46	67.08	69.00	35.89	56.94	58.94	88.37	44.72

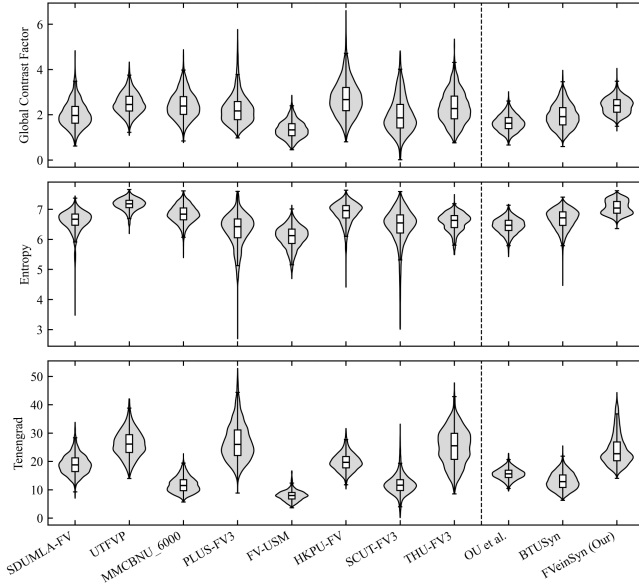


Fig. 7. Global contrast factor, Entropy, and Tenengrad gradient distributions of image quality metrics for real and synthetic finger vein datasets.

interest. In general, clearer vein structures are associated with higher entropy values. From this perspective, the entropy distribution of FVeinSyn-generated images is also highly consistent with those of the UTFVP and MNCBNU_6000 datasets. In addition, Tenengrad measures image sharpness by quantifying gradient strength, where higher values indicate clearer structural edges and richer details. Taken together, these results show that the proposed FVeinSyn method achieves statistical distributions on key metrics, including GCF, entropy, and Tenengrad gradient, that are most closely aligned with high-quality real datasets, while exhibiting overall superior performance compared to other synthetic approaches. This suggests that FVeinSyn-generated finger vein images demonstrate high authenticity and naturalness in terms of contrast, information representation, and clarity, thereby validating the effectiveness of the proposed method in modeling realistic pattern and overall visual quality.

D. CrowdSourced Evaluation

To assess the realism of the samples generated by FVeinSyn, we conducted a crowdsourced study involving ten researchers specializing in finger vein recognition. The evaluation focused on two aspects: (1) comparing the perceptual realism of our

TABLE III
AVERAGE REALISM RATINGS OF DIFFERENT SYNTHESIS METHODS (9 = MOST REALISTIC, 1 = LEAST REALISTIC)

Method	Region	Avg. Realism Rating	Std. Dev.
BTUSyn [37]	ROI	3.74	0.60
Ou <i>et al.</i> [38]	ROI	4.61	0.78
FVeinSyn (Our)	ROI	5.65	0.53
FVeinSyn (Our)	Full Finger	7.11	0.56

TABLE IV
PERCENTAGE CLASSIFIED AS REAL PER SYNTHESIS DATA

Method	Region	Average Rating (%)	Std. Dev. (%)
BTUSyn [37]	ROI	20.50	14.80
Ou <i>et al.</i> [38]	ROI	40.84	16.23
FVeinSyn (Our)	ROI	82.97	10.98
FVeinSyn (Our)	Full Finger	90.54	8.03

synthetic images with real data and several representative synthesis methods, and (2) measuring the probability of synthetic images were perceived as real. Evaluation samples were drawn from eight public datasets, two baseline synthesis approaches (Ou *et al.* [38] and BTUSyn [37]), and the proposed FVeinSyn framework. Both full-finger and ROI image types were assessed independently.

For the realism scoring task, observers were shown groups of nine images containing a mix of real and synthesized samples. Each image was rated on a 1–9 scale, where 9 indicates the highest realism. Real and synthetic samples were balanced within each group, and both the order and composition were randomized. Each participant evaluated at least 20 groups. Table III summarizes the mean scores and standard deviations. FVeinSyn obtained the highest realism ratings for both full-finger and ROI images. Traditional modeling-based synthesis (BTUSyn [37]) exhibited noticeably lower realism, while learning-based methods (Ou *et al.* [38]) produced improved results but still suffered from local artifacts arising from limited supervision.

In the real–fake discrimination task, observers were shown at least 90 images in random order and asked to decide whether each was “real” or “synthetic.” Table IV reports the proportion of images from each method that were judged as real. FVeinSyn again achieved the highest perceived-real rate, reaching $90.54 \pm 8.03\%$ for full-finger images and $82.97 \pm 10.98\%$ for ROI images. These results collectively demonstrate that FVeinSyn produces synthetic samples with perceptual realism and naturalness closest to real data.

TABLE V
PERFORMANCE COMPARISON OF VARIOUS ENHANCEMENT OR SYNTHESIS METHODS FOR FINGER VEIN RECOGNITION

Method	Configs			Performance (TAR@FAR=1e-3) $\uparrow$								
	#IDs	#per ID	#Image	SDUMLA-FV	UTFVP	MMCBNU_6000	PLUS-FV3	FV-USM	HKPU-FV	SCUT-FV3	THU-FV3	Avg.
MC	-	-	-	0.3117	0.5593	0.7669	0.6895	0.7632	0.3066	0.6261	0.5016	0.5656
RLT	-	-	-	0.4006	0.7093	0.7650	0.5363	0.8406	0.4753	0.5853	0.5788	0.6114
IUWT	-	-	-	0.4273	0.8269	0.7951	0.6190	0.7955	0.5758	0.5775	0.5405	0.6447
GF	-	-	-	0.3820	0.7232	0.8640	0.5104	0.8585	0.4743	0.5721	0.6158	0.6250
Real data	1.9k	-	18k	0.4602	0.7593	0.8385	0.5385	0.7709	0.6447	0.5504	0.5799	0.6428
Ou <i>et al.</i>	3k	1	3k	0.5692	0.6852	0.8239	0.5004	0.7340	0.6223	0.5283	0.5118	0.6219
BTUSyn	1.6k	10	16k	0.6262	0.8222	0.8383	0.4614	0.7856	0.6938	0.5478	0.5772	0.6691
FVeinSyn	1.9k	10	19k	0.7155	0.9222	0.9169	0.6852	0.8859	0.8903	0.5729	0.7922	0.7976
FVeinSyn	1.9k	50	95k	0.7941	0.9546	0.9438	0.8156	0.9305	0.9203	0.6532	0.8875	0.8625
FVeinSyn	10k	10	100k	0.8031	0.9500	0.9457	0.7935	0.9098	0.9295	0.6574	0.8714	0.8576
FVeinSyn	10k	50	500k	0.9402	0.9778	0.9538	0.9132	0.9551	0.9655	0.7078	0.9233	0.9171

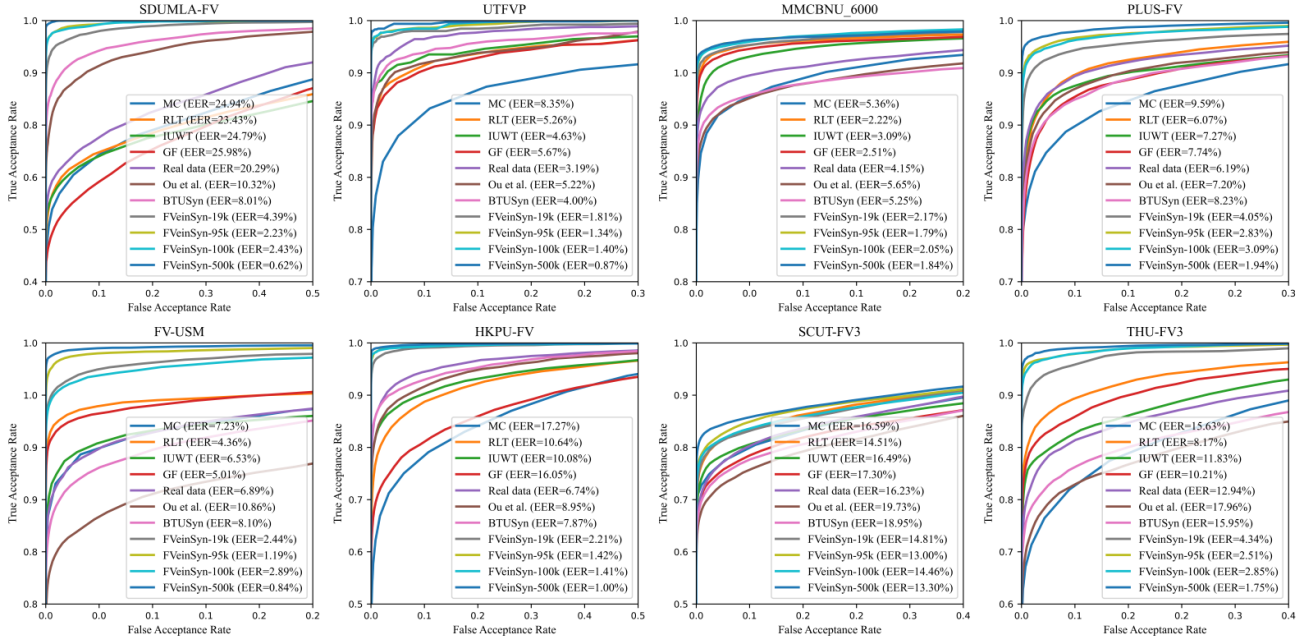


Fig. 8. Comparison of receiver operating characteristic (ROC) curves and equal error rates (EERs) for FVeinSyn and other representative methods under the open-set finger vein recognition protocol.

E. Recognition Performance Evaluation

Open-Set Finger Vein Recognition: Under the open-set protocol, we conducted a systematic comparison between real and synthetic data for finger vein recognition. The experiments were designed with a strict non-overlapping identity partition, ensuring that the vein identities in the training and test sets are completely independent. We compared the proposed FVeinSyn with representative traditional pattern-based matching methods [27], [37], [42]–[44] and learning-based approaches [38]. The recognition performance across multiple datasets is summarized in Table V. The results demonstrate that synthetic finger vein images generated by FVeinSyn exhibit a clear advantage: under the same number of identities and sample scale, the recognition performance improves by an average of 15.48% compared with models trained on real data. When the training set size and sample diversity are further increased, the improvement reaches up to 27.43%. Compared with other finger vein synthesis methods, FVeinSyn consistently achieves

higher recognition accuracy and more stable performance across all datasets. Fig. 8 presents the ROC curves and equal error rates (EERs) of different methods on various datasets, where FVeinSyn attains the largest area under the curve (AUC) and the lowest EER. Results show that FVeinSyn effectively improving the performance and robustness of recognition systems trained with synthetic data.

Cross-Domain Recognition: Cross-domain performance remains a critical challenge for practical finger vein recognition systems. We evaluated models trained with either real or synthetic data and tested their performance across different domains. The quantitative results are summarized in Table VI. Experimental findings show that pretraining with FVeinSyn-generated data significantly enhances cross-domain generalization, achieving an average recognition improvement of 40.87% per dataset and over 50% on several benchmarks. The largest improvement, 61.44%, is observed on the HKPU-FV dataset, which exhibits notable differences from other datasets,

TABLE VI

CROSS-DATASET EVALUATION USING OR NOT USING FVEINSYN FOR PRE-TRAINING. ✓ INDICATES THAT PRE-TRAINING WITH SYNTHETIC DATA IS USED (TAR@FAR=1E-3) ↑

Test Train	SDUMLA_FV		UTFVP		MMCBNU_6000		PLUS-FV3		FV-USM		HKPU-FV		SCUT-FV3		THU-FV3	
PT w/ Syn	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓
SDUMLA_FV	–	–	0.4065	0.8727	0.5565	0.9158	0.1344	0.7164	0.3117	0.7876	0.1317	0.7426	0.0120	0.4293	0.4464	0.6962
UTFVP	0.5083	0.8007	–	–	0.5850	0.8953	0.1611	0.7738	0.2421	0.7654	0.1375	0.8118	0.0281	0.4601	0.4834	0.6973
MMCBNU_6000	0.5137	0.8230	0.2704	0.8889	–	–	0.2095	0.7977	0.3787	0.8672	0.2127	0.8203	0.0334	0.5536	0.4572	0.6615
PLUS-FV3	0.5362	0.8023	0.3167	0.8819	0.4953	0.8401	–	–	0.1954	0.8375	0.0994	0.6554	0.0024	0.2370	0.4583	0.6193
FV-USM	0.4822	0.8019	0.2093	0.8991	0.7940	0.9372	0.2323	0.8375	–	–	0.2284	0.8547	0.1225	0.6056	0.4387	0.6821
HKPU-FV	0.5973	0.8626	0.4630	0.9509	0.7570	0.9264	0.1874	0.8040	0.4162	0.8040	–	–	0.1538	0.5349	0.5000	0.6699
SCUT-FV3	0.5438	0.7999	0.4625	0.9560	0.7346	0.9471	0.3159	0.6582	0.6913	0.8983	0.2770	0.8643	–	–	0.4806	0.6787
THU-FV3	0.5520	0.8168	0.5264	0.7444	0.7396	0.9355	0.2333	0.7475	0.3318	0.8323	0.1897	0.8066	0.0536	0.6485	–	–
Avg.	0.5334	0.8153	0.3793	0.8848	0.6660	0.9139	0.2106	0.7622	0.3667	0.8275	0.1823	0.7937	0.0580	0.4956	0.4664	0.6721
Imp.	↗	0.2819	↗	0.5056	↗	0.2479	↗	0.5516	↗	0.4608	↗	0.6114	↗	0.4376	↗	0.2058

TABLE VII

PERFORMANCE COMPARISON WITH INCREASING TRAINING SAMPLES (TAR@FAR=1E-3) ↑

Method \ #IDs	50	100	200	500	1000	1969
Real Data	0.0008	0.1765	0.4071	0.4926	0.5577	0.6222
Ou <i>et al.</i> [38]	0.0870	0.3883	0.4741	0.5310	0.5726	0.6038
BTUSyn [37]	0.1753	0.3890	0.4801	0.5296	0.5775	0.6487
FVeinSyn (Our)	0.3148	0.6924	0.8095	0.8142	0.8600	0.8748

such as brighter backgrounds and distinct finger regions. The poor cross-domain performance of models trained solely on real data primarily stems from limited dataset size and domain bias introduced by varying imaging conditions. These constraints hinder the model's ability to generalize across variations in illumination, background, and finger positioning. By synthesizing samples with diverse inter-class and cross-domain variations, FVeinSyn effectively addresses this limitation, enabling the model to learn more comprehensive and robust feature representations, thereby achieving substantially improved cross-domain recognition performance.

Identify Limited Recognition: We consider an identity-limited open-set recognition scenario to compare the performance of different finger vein synthesis methods. Using a 1:1 open-set protocol, we evaluate recognition performance as the number of training identities gradually increases (50, 100, 200, 500, 1000, and 1969), as summarized in Table VII.

The results show that when the number of training identities is extremely small (50 IDs), both Ou *et al.* [38] and BTUSyn [37] exhibit limited recognition performance, failing to ensure model stability. In contrast, models pretrained with FVeinSyn-generated images maintain strong recognition performance even under such highly constrained conditions. Notably, when only 100 training identities are used (5% of the full set), the FVeinSyn-pretrained model still outperforms the baseline model trained on all 1,969 real identities. These findings indicate that FVeinSyn effectively enhances inter-class diversity and discriminative feature learning under limited-identity conditions, thereby improving model generalization and open-set recognition performance.

Sample Limited (One/Few Shot) Recognition: We consider

a more practical recognition scenario—sample-limited recognition, users typically provide only a very small number of samples during registration, and sometimes even a single sample. Under this condition, we evaluated recognition models trained with different synthetic data, and further fine-tuned them with N enrollment samples ($N = 1, 2, 3$). In the experiments, FVeinSyn was systematically compared with real data and two representative finger vein synthesis methods. The results are summarized in Table VIII. Fig. 9 presents the ROC curves and EERs of different sample configurations on various datasets, where FVeinSyn attains the largest area under the curve and the lowest EER.

As shown in the table, models trained solely with a limited number of real samples generally exhibit low recognition performance, making it difficult to ensure system reliability. The introduction of synthetic data alleviates this problem to some extent, with BTUSyn showing moderate improvement over traditional approaches, yet its overall performance remains constrained. In contrast, pretraining with FVeinSyn-generated data followed by fine-tuning with a few real samples leads to a substantial performance improvement. Across all datasets and sample configurations, FVeinSyn consistently achieves the highest recognition accuracy and maintains strong generalization even under the extreme single-sample condition ($N = 1$). As the number of enrollment samples increases from $N = 1$ to $N = 3$, all methods benefit from performance gains, but the improvement obtained by FVeinSyn is the most pronounced. This indicates that the samples synthesized by FVeinSyn better capture inter-class variability and structural consistency, resulting in more representative and learnable features.

F. Synthetic Dataset Evaluation

In this section, we follow the three dependency-type metrics [34] to evaluate synthetic datasets to help us understand the properties of the generated datasets. We generated 10,000 vein identities using FVeinSyn, with 50 samples per identity, total of 500,000 finger vein images. F_{eval} is the recognition model used to evaluate synthetic finger vein datasets. The more generalizable F_{eval} is, the more accurate the metrics are in terms of the identity and diversity of the generated synthetic

TABLE VIII
PERFORMANCE COMPARISON OF ONE/FEW-SHOT FINGER VEIN RECOGNITION (TAR@FAR=1E-3) $\uparrow$

	Dataset	SDUMLA-FV	UTFVP	MMCBNU_6000	PLUS-FV3	FV-USM	HKPU-FV	SCUT-FV3	THU-FV3
$N = 1$	Real data	0.4736	0.2815	0.6874	0.2655	0.4365	0.2359	0.1034	0.4902
	Ou <i>et al.</i> [38]	0.5319	0.3380	0.7338	0.3928	0.4459	0.2649	0.0576	0.5628
	BTUSyn [37]	0.5855	0.5333	0.8376	0.3834	0.6125	0.3849	0.1023	0.6213
	FVeinSyn (Our)	0.7220	0.7315	0.8710	0.8874	0.5650	0.8096	0.2626	0.5826
$N = 2$	Real data	0.5440	0.4333	0.7815	0.3823	0.6490	0.3889	0.4070	0.5681
	Ou <i>et al.</i> [38]	0.5810	0.4583	0.8184	0.5367	0.5384	0.3462	0.3030	0.6123
	BTUSyn [37]	0.6475	0.7139	0.8828	0.5216	0.6930	0.4723	0.3722	0.6973
	FVeinSyn (Our)	0.8913	0.9194	0.9645	0.9590	0.9392	0.9643	0.7491	0.9394
$N = 3$	Real data	0.5975	—	0.7784	0.5318	0.7231	0.5196	0.5061	0.6202
	Ou <i>et al.</i> [38]	0.6641	—	0.8513	0.6078	0.6456	0.4418	0.3554	0.7621
	BTUSyn [37]	0.7112	—	0.9020	0.6042	0.7640	0.5826	0.4189	0.8253
	FVeinSyn (Our)	0.9497	—	0.9786	0.9878	0.9779	0.9868	0.8835	0.9793

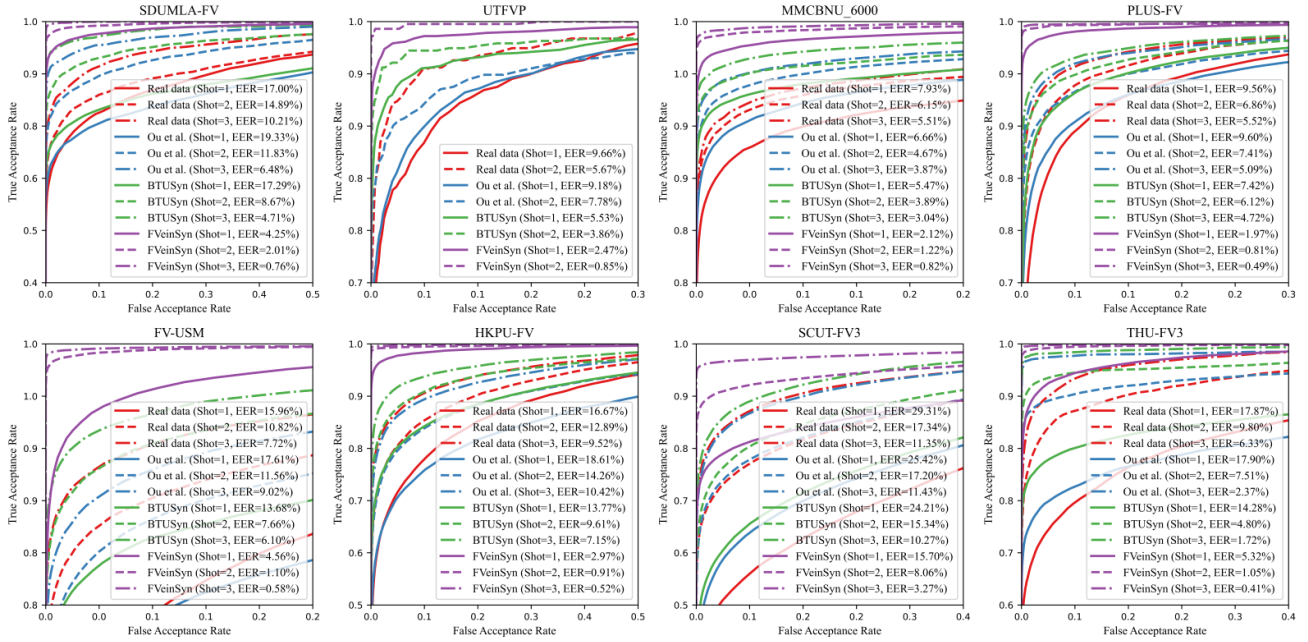


Fig. 9. Comparison of receiver operating characteristic (ROC) curves and equal error rates (EERs) for FVeinSyn and other representative methods under the limited finger vein sample recognition protocol.

datasets. Let y_c be the class label, and $f_i = F_{eval}(x_i)$. $d(f_i, f_j)$ represents the distance between two images in the F_{eval} feature space.

Uniqueness: The uniqueness U_{class} can be used to quantify the number of distinct identities generated in a synthetic dataset. During evaluation, we determine the capacity limit of the synthetic dataset U by constructing non-overlapping r -ball regions in the F_{eval} feature space—continuing to add new regions until no further additions can be made without overlap. This process is constrained by two key parameters: the feature distance threshold r (used to determine identity matching) and the feature extraction capability of the F_{eval} model.

In implementation, we first compute the mean feature vector (cluster center) for all samples belonging to the same identity. If the distance between two cluster centers exceeds threshold r , they are classified as distinct identities. The uniqueness metric U_{class} is then defined as $\frac{|U_c|}{C}$, where the numerator $|U_c|$ represents the actual count of unique identities generated in the synthetic dataset, and the denominator C denotes the

total number of predefined identity classes in F_{eval} . As shown in Table IX, the identity distinctiveness of synthetic dataset is 99.83%, indicating that the generated finger vein patterns exhibit significant differences. Only a very small portion of the finger vein identity feature vectors show low similarity ($r < 0.2$).

Intra-Class Consistency: C_{intra} is used to evaluate feature coherence among samples belonging to the same unique identity, quantified by measuring the distance between each sample and its corresponding identity feature center. This experiment uses the bounding box annotation provided by the synthetic dataset to extract finger vein ROI images for recognition. Benefiting from the accurate annotation, the synthetic dataset has an intra-class consistency C_{intra} of 100%.

Intra-Class Diversity: D_{intra} measures the variation among generated samples within the same identity. Given the inherently limited intra-class variations in existing public finger vein datasets, we cannot directly adopt the method from [34]. Instead, we train a recognition model F_n using aligned and

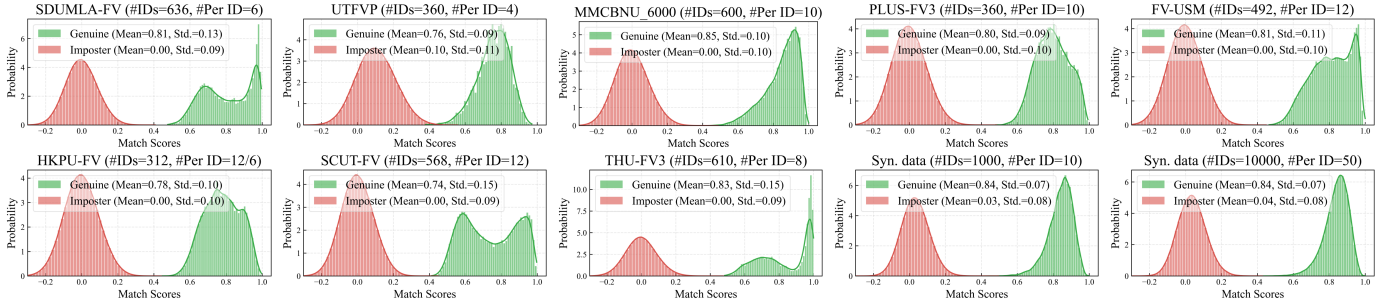


Fig. 10. Genuine/Impostor matching score distributions across multiple real datasets and the FVeinSyn synthetic dataset.

TABLE IX
DATASET EVALUATION OF SYNTHETIC FINGER VEIN

Dataset	U_{class}	C_{intra}	D_{intra}
Ou <i>et al.</i> [38]	24.57	—	—
BTUSyn [37]	19.19	99.98	3.28
FVeinSyn (Our)	99.83	100.00	89.67

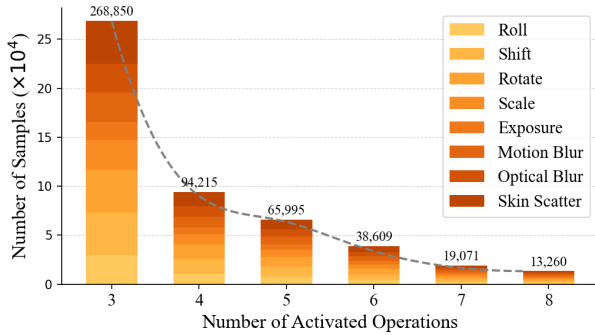


Fig. 11. Distribution of the number of intra-class variations in 500,000 FVeinSyn generated samples.

normalized samples. In this feature space, two samples from the same identity are considered distinct if their distance exceeds threshold r . The metric is formally defined as

$$D_{intra} = \frac{1}{C} \frac{2}{N(N-1)} \sum_{c=1}^C \sum_{i=1}^N \sum_{j=i+1}^N \delta[d(f_{n_i}^c, f_{n_j}^c) < r], \quad (23)$$

where f_n^c is the n -th sample of class c , and $\delta[\text{condition}] = 1$ if the condition is true, otherwise it is 0. D_{intra} ranges from 0 to 1, with higher values indicating greater intra-class variation. As shown in Table IX, the intra-class diversity of synthetic dataset is 89.67%.

We plotted the matching score distributions of eight real finger vein datasets and the FVeinSyn generated synthetic dataset under varying numbers of identities and sample sizes, as shown in Fig. 10. The results indicate that the synthetic and real datasets exhibit high consistency in key statistical measures, such as mean and variance, demonstrating significant statistical alignment. Specifically, the synthetic data closely matches the real data: the mean genuine/impostor matching scores of the real datasets range from 0.74–0.85/0.00–0.10, while the synthetic dataset shows 0.84/0.03–0.04, well within the typical range of real data; for variance, the real datasets’

genuine/impostor scores span 0.09–0.15/0.09–0.11, and the synthetic data yields 0.07/0.08, again demonstrating strong alignment with real distributions. This further confirms the reliability of FVeinSyn generated synthetic data in terms of both realism and diversity.

G. Ablation Study on Intra-Class Diversity Generator

We conduct an ablation study to examine the contribution of different intra-class variation operators in the diversity generator. Specifically, we construct synthetic pretraining datasets with isolated transformations, including geometric perturbations and optical degradations, and pretrain separate models for evaluation under the same setting as open-set recognition.

As shown in Table X, all individual transformations improve recognition performance over the real-data baseline, indicating that both geometric and optical variations provide useful supervision during pretraining. These two categories contribute in a complementary manner, with geometric perturbations introducing structural variability and optical degradations modeling appearance-level changes.

The operator weights used in the diversity generator are consistent with the relative contributions observed in this ablation study. Specifically, shift, rotation, and skin scattering are assigned higher weights, while exposure is assigned a lower weight. The superior performance of the mixed setting further supports the use of empirically weighting for jointly modeling structural and appearance-level variations.

Furthermore, we analyze the distribution of intra-class variations in the generated samples. As shown in Fig. 11, the number of variations per sample follows a long-tail distribution, where most samples contain only a few variations while samples with more variations gradually decrease. This trend is consistent with real-world data, indicating that the proposed diversity generator not only improves recognition performance but also produces samples with realistic statistical properties.

H. Demographic Bias Analysis

The training of the renderer depends on real data, the generated data may also inherit or amplify potential demographic biases in real data. This issue is particularly important for synthetic finger-vein data used as a pretraining foundation, because once unrecognized bias exists in the generation process, it may further affect the generalization performance of downstream recognition models across different demographic

TABLE X
IMPACT OF DIFFERENT INTRA-CLASS VARIATIONS ON RECOGNITION PERFORMANCE ACROSS DATASETS (TAR@FAR=10⁻³) $\uparrow$.

Intra-Class Variation	SDUMLA-FV	UTFVP	MMCBNU_60000	PLUS-FV3	FV-USM	HKPU-FV	SCUT-FV	THU-FV3	Avg.
Real data	0.4602	0.7593	0.8385	0.5385	0.7709	0.6447	0.5504	0.5799	0.6428
FVeinSyn (Mixed)	0.9402	0.9778	0.9538	0.9132	0.9551	0.9655	0.7078	0.9233	0.9171
Geometric Perturbations									
Shift	0.8994	0.9528	0.9395	0.8749	0.9009	0.9264	0.6642	0.8736	0.8789
Rotation	0.9283	0.9528	0.9209	0.8086	0.8875	0.8850	0.6965	0.8371	0.8646
Rolling	0.9107	0.9509	0.8901	0.8619	0.8726	0.8545	0.6675	0.7963	0.8506
Scaling	0.9034	0.9463	0.8993	0.8628	0.8945	0.8971	0.5996	0.8326	0.8545
Optical Degradation									
Exposure	0.8805	0.9194	0.8510	0.7863	0.9012	0.8710	0.6498	0.7952	0.8318
Skin Scattering	0.9090	0.9546	0.9447	0.8791	0.9046	0.8965	0.6635	0.8885	0.8801
Motion Blur	0.8868	0.9435	0.8873	0.8400	0.8870	0.8757	0.6249	0.8409	0.8483
Optical Blur	0.9149	0.9306	0.8833	0.8209	0.8977	0.8647	0.6498	0.8389	0.8501

TABLE XI
DEMOGRAPHIC FAIRNESS ANALYSIS OF RECOGNITION PERFORMANCE WITH SYNTHETIC DATA (TAR@FAR=10⁻³) $\uparrow$.

Dataset	PT w/ Syn		Gender		Age			Race			Blood Type			
	$\times$	$\checkmark$	Male	Female	≤ 20	21-40	≥ 41	Asian	Caucasian	African	A	B	AB	O
UTFVP	0.7593	0.9778	0.9822	0.9222	0.9815	0.9759	0.9514	-	-	-	0.9407	0.9874	0.9704	0.9663
MMCBNU_6000	0.8385	0.9538	0.9458	0.9876	1.0000	0.9494	0.9605	0.9542	0.9531	0.9741	0.9407	0.9874	0.9704	0.9663

groups. Therefore, it is necessary to analyze the impact of pre-training with data synthesized by FVeinSyn on the recognition performance of different demographic groups.

Specifically, UTFVP and MMCBNU_6000, two evaluation datasets with available demographic metadata, are used for evaluation. Under the same open-set recognition setting as the main experiments, test identities are grouped according to different demographic attributes for subgroup evaluation. For UTFVP, the recognition results are reported for gender and age groups. For MMCBNU_6000, gender, age, race, and blood type are further analyzed. Table XI reports the overall performance with and without FVeinSyn pretraining, as well as the recognition results of each demographic subgroup after synthetic pretraining, which are used to compare the differences between in-group performance and overall performance.

From the results of each demographic subgroup, it can be observed that the performance of different gender, age, race, and blood type groups is generally close to the overall performance, and the differences between subgroup performance and overall performance remain small. No specific group shows a persistent and significant deviation from the overall trend. At the same time, the performance gains brought by FVeinSyn pretraining are relatively stable across different groups, without being clearly concentrated on any particular group. No significant disproportionate performance degradation is observed for underrepresented groups.

I. Computational Efficiency

FVeinSyn adopts a three-stage generation pipeline, and the total time required to synthesize a finger vein image is the sum of the times consumed by each stage, as summarized in Table XII. Specifically, the process consists of three steps: i) Identity Generation (285.6 ms), ii) Render (7.9 ms), and iii) Diversity Generation (0.7 ms). Consequently, the total time to generate a single synthetic sample is approximately 294.2 ms.

TABLE XII
TIME CONSUMPTION OF FINGER VEIN IMAGE SAMPLE USING FVEIN SYN

Step	Identity Generation	Render	Diversity Generation	Total
Time (ms)	285.6	7.9	0.7	294.2

Each generated sample includes a 300 $\times$ 600 8-bit grayscale finger vein image, which requires about 55 KB storage in loss-less Portable Network Graphics (PNG) format, an associated XML annotation file (approximately 2 KB), and a pixel-wise annotation mask representing the finger shape, vein pattern, and joint cavity regions (approximately 4 KB). FVeinSyn supports offline generation of synthetic finger vein samples for training, as well as on-the-fly generation during training, thereby effectively reducing the storage overhead.

V. CONCLUSION

In this work, we presented FVeinSyn, a large-scale controllable synthetic data generation framework for finger vein recognition. By explicitly decoupling vascular topology synthesis from imaging appearance rendering, FVeinSyn effectively addresses the limitations caused by insufficient training data, such as low identity diversity and limited image realism. Our framework incorporates three collaborative modules: a finger vein identity generator to produce anatomically valid and identity-distinctive vein patterns, a cascaded region-aware GAN for realistic near-infrared rendering, and an intra-class diversity generator to simulate realistic intra-class variations. We used FVeinSyn to generate a dataset of 500,000 images across 10,000 vein identities and conducted extensive evaluations. Experimental results demonstrated that models trained with FVeinSyn achieve superior recognition performance, enhanced cross-domain robustness, and improved performance in data-limited scenarios, with an average recognition performance gain of 27.43% compared to real-data baselines.

These results confirm that FVeinSyn not only produces highly realistic and diverse synthetic samples but also significantly benefits downstream finger vein recognition tasks.

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